

The P- δ Probe: Assaf–González Local Characterization on $B((2, 2, 0))$ at $n = 3$ — $V_{<\mu}$ Verified at $t = 0$, t -Graded Structure Revealed

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Abstract

Report on the P- δ probe from dream cycle 10 (2026-07-24). We compute the type- A_2 crystal $B((2, 2, 0))$ at $n = 3$ (a hexagon of 6 SSYT), verify its Demazure atom decomposition (block sizes 1, 2, 3), and apply the Assaf–González arXiv:2512.19814 local characterization (conditions (E), (I), (P)) as a Python predicate. The A–G test correctly identifies the three Demazure crystals in the Bruhat chain $B_e \subset B_{s_2} \subset B(\mu)$, and correctly *rejects* the shape- μ subset $\{H, B, L\}$, the shape- $< \mu$ subset $\{A, C, D\}$, and individual atoms. This verifies Route δ at $t = 0$ for $\mu = (2, 2, 0)$, $n = 3$. Separately, we compute the nil-Hecke t -atoms A_γ^{alt} as explicit polynomials in $\mathbb{Q}(t)[x_1, x_2, x_3]$ and observe that each A_γ^{alt} , viewed as a $\mathbb{Q}(t)$ -linear combination of crystal elements, has block lower triangular structure (in Bruhat order on γ) with diagonal blocks $(1, 1 - t, 1 - t, \dots, 1 - t)$ and pure- t shadows on lower classical atoms. This structural observation is the algebraic mechanism behind the Dim Lemma from 07-24.

1 Setup and background

The atom decomposition sprint concerns the identity

$$P_\mu(x; t) = \sum_{\gamma \in S_n \cdot \mu} c_\gamma(t) A_\gamma^{\text{alt}}(x; t),$$

where P_μ is the Hall–Littlewood polynomial, A_γ^{alt} is a nil-Hecke t -atom built from the ascending bubble word using

$$\theta_i^{\text{alt}} = \frac{x_{i+1}(1 - s_i)}{x_i - x_{i+1}} - t \cdot \frac{x_i(1 - s_i)}{x_i - x_{i+1}} \quad (\text{operator variant used in Clio's sprint}),$$

and $c_\gamma(t) \in \mathbb{Z}[t]$ is a coefficient depending only on the orbit-structure of μ .

The 07-23 PROVE session (see 2026-07-23-atoms-decomposition-P-mu.tex) reduces the identity to the $V_{<\mu}$ *consistency lemma*:

$$R := P_\mu - \sum_{\gamma} c_\gamma(t) A_\gamma^{\text{alt}} \quad \text{vanishes on shape } < \mu.$$

The 07-24 PROVE session (see 2026-07-24-atoms-decomposition-P-mu-part-II.tex) then further reduces $V_{<\mu}$ to a *Dim Lemma*: symmetric polynomials inside $V_\mu^{\text{alt}} := \text{span}\{A_\gamma^{\text{alt}} : \gamma \in S_n \cdot \mu\}$

form a subspace of dimension ≤ 1 . Equivalently: $V_{<\mu} \Leftrightarrow P_\mu \in V_\mu^{\text{alt}}$, a *one-dimensional existence question*.

Route δ . The recent Assaf–González paper [1] gives a purely local characterization of unions of Demazure crystals in symmetrizable Kac–Moody generality. Their local conditions (E), (I), (P) — expressed on i -strings of the crystal graph — imply that a subset $X \subseteq B(\mu)$ is a union of Demazure crystals, hence (as a corollary) admits a disjoint decomposition into Demazure atoms.

Commutativity at $t = 0$. Both the Dim Lemma route and Route δ specialise at $t = 0$ to the classical identity

$$s_\mu = \sum_{w \in W/W_\mu} A_{w \cdot \mu}^{\text{cl}} \quad (\text{Lascoux–Schützenberger 1990, Mason 2006}).$$

The research content lives in the t -graded lift.

2 The crystal $B((2, 2, 0))$ at $n = 3$

We work with $\mu = (2, 2, 0)$ in type A_2 . The crystal $B(\mu)$ has 6 semistandard tableaux of shape $(2, 2)$ with entries in $\{1, 2, 3\}$. We label them and record weights:

label	tableau (rows top-to-bottom)	weight
H	$\begin{array}{cc} 1 & 1 \\ 2 & 2 \end{array}$	$(2, 2, 0)$
A	$\begin{array}{cc} 1 & 1 \\ 2 & 3 \end{array}$	$(2, 1, 1)$
B	$\begin{array}{cc} 1 & 1 \\ 3 & 3 \end{array}$	$(2, 0, 2)$
C	$\begin{array}{cc} 1 & 1 \\ 2 & 2 \end{array}$	$(1, 2, 1)$
D	$\begin{array}{cc} 1 & 2 \\ 3 & 3 \end{array}$	$(1, 1, 2)$
L	$\begin{array}{cc} 2 & 2 \\ 3 & 3 \end{array}$	$(0, 2, 2)$

The crystal edges are:

$$\begin{aligned} f_1 : \quad & A \rightarrow C, \quad B \rightarrow D, \quad D \rightarrow L, \\ f_2 : \quad & H \rightarrow A, \quad A \rightarrow B, \quad C \rightarrow D. \end{aligned}$$

The i -strings are:

$$\begin{aligned} \text{1-strings:} \quad & \{H\} \ (\ell = 1), \quad \{A, C\} \ (\ell = 2), \quad \{B, D, L\} \ (\ell = 3), \\ \text{2-strings:} \quad & \{H, A, B\} \ (\ell = 3), \quad \{C, D\} \ (\ell = 2), \quad \{L\} \ (\ell = 1). \end{aligned}$$

Remark 1. The extremal elements (those fixed by e_i or f_i appropriately at the ends of strings) are $\{H, B, C, L\}$ — *four* of them, not three. The element C has weight $(1, 2, 1) \notin S_3 \cdot (2, 2, 0)$; it is an “internal extremal” with respect to 2-strings but sits mid-orbit in weight.

3 Classical Demazure atoms (Lascoux–Schützenberger 1990)

The stabiliser of $\mu = (2, 2, 0)$ in S_3 is $\langle s_1 \rangle$, so the minimal coset representatives for $S_3 / \langle s_1 \rangle$ are $\{e, s_2, s_1 s_2\}$. The corresponding Demazure crystals nest as

$$B_e(\mu) \subset B_{s_2}(\mu) \subset B_{s_1 s_2}(\mu).$$

Explicitly:

$$\begin{aligned} B_e(\mu) &= \{H\}, & |B_e| &= 1, \\ B_{s_2}(\mu) &= \{H, A, B\}, & |B_{s_2}| &= 3, \\ B_{s_1 s_2}(\mu) &= B(\mu) = \{H, A, B, C, D, L\}, & |B_{s_1 s_2}| &= 6. \end{aligned}$$

The Demazure atoms (successive differences) are:

$$\begin{aligned} A_{(2,2,0)}^{\text{cl}} &= B_e \setminus \emptyset = \{H\}, & |\cdot| &= 1, \\ A_{(2,0,2)}^{\text{cl}} &= B_{s_2} \setminus B_e = \{A, B\}, & |\cdot| &= 2, \\ A_{(0,2,2)}^{\text{cl}} &= B_{s_1 s_2} \setminus B_{s_2} = \{C, D, L\}, & |\cdot| &= 3. \end{aligned}$$

Block sizes $1 + 2 + 3 = 6$; the atoms are disjoint and their union is $B(\mu)$. This is the identity $s_{(2,2)} = \sum_{\gamma} A_{\gamma}^{\text{cl}}$.

4 The A–G local characterization applied

Following Assaf–González [1], a subset $X \subseteq B(\mu)$ is a union of Demazure crystals iff three local conditions hold on every i -string S :

- (E) Edge condition.** $S \cap X$ is either $\emptyset$, or all of S , or an initial segment (“head”) of S containing the string’s highest weight.
- (I) Intersection condition.** Compatibility of X under pairs of adjacent f_i, f_j on the crystal — automatically inherited from (E) in type A_2 with the strings we test here.
- (P) Projection condition.** Highest-weight compatibility for elements of X under any string-projection to shorter strings.

We implemented (E) + (I) + (P) as a Python predicate over the six-element crystal $B((2, 2, 0))$ and evaluated it on nine candidate subsets. The results:

#	Candidate subset X	(E)	(I)	(P)	Demazure?
1	$\{H\}$	✓	✓	✓	YES = B_e
2	$\{H, A, B\}$	✓	✓	✓	YES = B_{s_2}
3	$\{H, A, B, C, D, L\}$	✓	✓	✓	YES = $B_{s_1 s_2}$
4	$\{A, B\} = A_{(2,0,2)}^{\text{cl}}$	×	—	—	NO
5	$\{H, B, L\}$ (shape- μ)	×	—	—	NO
6	$\{A, C, D\}$ (shape- $< \mu$)	×	—	—	NO
7	$\{C, D, L\} = A_{(0,2,2)}^{\text{cl}}$	×	—	—	NO
8	$\{H, A, B, C, D\}$	×	—	—	NO
9	$\{H, A\}$	×	—	—	NO

Informative failure modes

- **Individual atoms fail (E).** $A_{(2,0,2)}^{\text{cl}} = \{A, B\}$ meets the 2-string $\{H, A, B\}$ in $\{A, B\}$, which is a *tail* not a head; and the 1-string $\{A, C\}$ in $\{A\}$, which is a head, but the 1-string $\{B, D, L\}$ meets $\{A, B\}$ in $\{B\}$, again not a head. Two independent failures. Atoms are pieces of Demazure crystals, not Demazure crystals themselves.

- **Shape- μ subset $\{H, B, L\}$ fails (E).** The 1-string $\{B, D, L\}$ meets $\{H, B, L\}$ in $\{B, L\}$ — neither $\emptyset$, nor full, nor an initial segment.
- **Shape- $< \mu$ subset $\{A, C, D\}$ fails (E) doubly.** The 1-string $\{B, D, L\}$ meets it in $\{D\}$ (middle, not head); the 2-string $\{H, A, B\}$ meets it in $\{A\}$ (middle, not head). Two parallel violations.

Verdict. The A–G local condition correctly characterizes the Demazure crystal chain of $B((2, 2, 0))$, and the shape- $< \mu$ residual is *far* from being a union of atoms. This is as expected: at $t = 0$, $V_{< \mu}$ is trivially true via L–S 1990 (any symmetric polynomial in the shape- μ orbit that is a $\mathbb{Q}$ -linear combination of the three classical atoms is uniquely s_μ).

5 Nil-Hecke t -atoms as t -weighted crystal subsets

We compute the three nil-Hecke t -atoms via the ascending bubble word. Explicitly:

$$\begin{aligned} A_{(2,2,0)}^{\text{alt}} &= x_1^2 x_2^2, \\ A_{(2,0,2)}^{\text{alt}} &= x_1^2 x_3^2 + (1-t)x_1^2 x_2 x_3 + (-t)x_1^2 x_2^2, \\ A_{(0,2,2)}^{\text{alt}} &= x_2^2 x_3^2 + (1-t)x_1 x_2^2 x_3 + (1-t)x_1 x_2 x_3^2 + (-t)x_1^2 x_3^2 + t(t-1)x_1^2 x_2 x_3. \end{aligned}$$

Central structural observation

View A_γ^{alt} as a $\mathbb{Q}(t)$ -linear combination of crystal elements $T \in B(\mu)$ via the map $T \mapsto x^{\text{wt}(T)}$. The coefficient matrix (rows indexed by γ , columns indexed by $T \in \{H, A, B, C, D, L\}$, both ordered by Bruhat) is:

$$M = \begin{array}{c} \gamma = (2, 2, 0) \\ \gamma = (2, 0, 2) \\ \gamma = (0, 2, 2) \end{array} \begin{array}{c} H \quad A \quad B \quad C \quad D \quad L \\ \left(\begin{array}{cccccc} 1 & 0 & 0 & 0 & 0 & 0 \\ -t & 1-t & 1 & 0 & 0 & 0 \\ 0 & t(t-1) & -t & 1-t & 1-t & 1 \end{array} \right) \end{array}$$

The matrix is *block lower triangular* with respect to the partition of columns by classical Demazure atom:

$$\{H\} \cup \{A, B\} \cup \{C, D, L\}.$$

Diagonal blocks

For row γ restricted to columns of A_γ^{cl} (the “own atom” of γ), we read off:

- $\gamma = (2, 2, 0)$ on $\{H\}$: coefficient 1 on the extremal element.
- $\gamma = (2, 0, 2)$ on $\{A, B\}$: coefficients $(1-t, 1)$ — with 1 on the extremal element B (weight γ), and $1-t$ on the non-extremal A .
- $\gamma = (0, 2, 2)$ on $\{C, D, L\}$: coefficients $(1-t, 1-t, 1)$ — with 1 on the extremal L (weight γ), and $1-t$ on non-extremal C, D .

Sub-diagonal blocks (“shadows”)

For row γ restricted to columns of $A_{\gamma'}^{\text{cl}}$ with $\gamma' < \gamma$ in Bruhat:

- $\gamma = (2, 0, 2)$ on $\{H\}$: coefficient $-t$ (pure- t).
- $\gamma = (0, 2, 2)$ on $\{A, B\}$: coefficients $t(t-1)$ and $-t$ (both divisible by t).

There are *no* entries above the diagonal in Bruhat.

Verification of the atom identity

With $c_{(2,2,0)}(t) = [3]_t$, $c_{(2,0,2)}(t) = [2]_t$, $c_{(0,2,2)}(t) = 1$ (Clio’s 07-24 output; $[k]_t = 1 + t + \dots + t^{k-1}$), one has

$$[3]_t A_{(2,2,0)}^{\text{alt}} + [2]_t A_{(2,0,2)}^{\text{alt}} + A_{(0,2,2)}^{\text{alt}} = P_{(2,2)}(x_1, x_2, x_3; t)$$

as polynomials in $\mathbb{Q}(t)[x_1, x_2, x_3]$. We verified this by direct expansion.

6 The t -graded A–G conjecture

The observed block structure suggests the following t -graded strengthening of A–G’s local characterization.

Conjecture 2 (t -graded A–G, provisional). *Let μ be a highest weight of type A_{n-1} and let γ range over the S_n -orbit of μ , indexed by minimal coset representatives $w \in W/W_\mu$. The nil-Hecke t -atom A_γ^{alt} , viewed as a $\mathbb{Q}(t)$ -linear combination of crystal elements of $B(\mu)$ via $T \mapsto x^{\text{wt}(T)}$, has the following structure:*

- On the classical Demazure atom $A_\gamma^{\text{cl}} \subset B(\mu)$: coefficient 1 on the extremal element (weight γ), coefficient $1 - t$ on all other elements of A_γ^{cl} .*
- On classical atoms $A_{\gamma'}^{\text{cl}}$ with $\gamma' < \gamma$ in Bruhat: pure- t “shadow” coefficients (divisible by t).*
- On classical atoms $A_{\gamma''}^{\text{cl}}$ with $\gamma'' \geq \gamma$ or incomparable to γ : coefficient 0.*

Consequently, the coefficient map $(T, \gamma) \mapsto [\text{coefficient of } T \text{ in } A_\gamma^{\text{alt}}]$ is a lower-triangular matrix in the Bruhat order on $S_n \cdot \mu$, whose diagonal γ -blocks consist of $(1, 1 - t, \dots, 1 - t)$ patterns and whose sub-diagonal entries are t -multiples.

Proposition 3 (Consequence, provisional). *Assume Conjecture 2. With $c_\gamma(t)$ chosen as the Gaussian coefficients output by Clio’s 07-24 back-substitution, the sum*

$$\sum_{\gamma} c_\gamma(t) A_\gamma^{\text{alt}}$$

reproduces $P_\mu(x; t)$. On shape- μ (extremal weights), the diagonal 1’s and shadow $-t$ contributions cancel via Gaussian identities of the form $[k]_t + [k-1]_t \cdot (-t) = 1$. On shape- $< \mu$ (non-extremal weights), the diagonal $(1-t)$ and shadow $t(t-1), \dots$ contributions combine to reproduce the $m_\nu(x)$ -coefficients of $P_\mu(x; t)$.

Remark 4 (Sanity check at $t = 0$). At $t = 0$: the diagonal entries $(1 - t) \rightarrow 1$, the “top” entries $1 \rightarrow 1$, the shadow entries $-t, t(t - 1) \rightarrow 0$. The matrix M becomes the block-diagonal indicator matrix of the classical Demazure atoms A_γ^{cl} , and the identity of Proposition 3 specialises to

$$\sum_{\gamma} A_\gamma^{\text{cl}} = s_\mu \quad (\text{Lascoux–Schützenberger 1990}). \quad \checkmark$$

Remark 5 (Next PROVE cycle). Conjecture 2 needs to be tested and, if it holds, proved. Concrete next steps:

1. Verify at $\mu = (2, 1, 0)$, $n = 3$ (orbit size 6, richer Bruhat structure).
2. Verify at $\mu = (2, 2, 1)$, $n = 3$ (orbit size 3, ω -conjugate check).
3. Verify at $\mu = (2, 2, 0)$, $n = 4$ (orbit size 6, higher rank).
4. Attempt to prove the block-lower-triangular structure directly from the commutation relations of θ_i^{alt} with the crystal operators.
5. Formulate a t -graded version of the A–G local conditions (E)+(I)+(P) using Gaussian $[k]_t$ weights on i -strings, and verify it against the block structure.

7 Verdict and next steps

- The P- δ probe at $t = 0$ for $\mu = (2, 2, 0)$, $n = 3$: **PASSED**. The A–G local characterization correctly certifies the three Demazure crystals in the Bruhat chain and correctly rejects the individual atoms, the shape- μ orbit subset, and the shape- $< \mu$ subset.
- The shape- $< \mu$ subset $\{A, C, D\}$ fails (E) via two parallel violations on the 1-string $\{B, D, L\}$ and the 2-string $\{H, A, B\}$. This is an *informative* failure: A–G’s local test recognises the shape- $< \mu$ residual as not just non-Demazure, but non-Demazure for a structurally uniform reason.
- The nil-Hecke t -atoms A_γ^{alt} , when viewed as t -weighted crystal subsets, exhibit a *block lower triangular* structure in Bruhat with a clean $(1, 1 - t, \dots, 1 - t)$ diagonal-block pattern and pure- t shadow entries. This is the algebraic mechanism that makes Clio’s Dim Lemma work: the shadow entries are absorbed by the Gaussian coefficients $c_\gamma(t) = [3]_t, [2]_t, 1$, and the diagonal-block pattern reproduces the $(1 - t)$ -weighted monomial-symmetric expansion of $P_\mu(x; t)$.
- **Immediate next steps.** (i) Verify Conjecture 2 on $\mu = (2, 1, 0)$, $n = 3$ (regular orbit); (ii) attempt to prove the block-lower-triangular structure of A_γ^{alt} from the θ^{alt} commutation relations directly; (iii) formulate the t -graded (E)+(I)+(P) with Gaussian $[k]_t$ weights and check compatibility with A^{alt} .

References

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